

The Immunity Factor from Contact Surveys: Where Epidemics Turn Over

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Abstract

Epidemics in 2020 grew as if the reproduction number were near 2.5 and then turned over with a fifth of the population infected, far below the herd-immunity threshold of 60% that the growth implies. Because the most active people are infected first, the effective reproduction number falls faster than the susceptible share S : as RS^λ rather than RS , where the immunity factor λ is one in a homogeneous population. The turnover is then $1 - R^{-1/\lambda}$. We measure λ from a contact panel that follows the same people for two years, which separates lasting differences in activity from passing ones, and obtain $\lambda \approx 3.5$ with nothing fitted to epidemic data. The same value is found in 341 US county waves of winter 2020–21 and in fits to the first wave in New York City, and it puts the turnover at $R = 2.5$ near 23%.

1 Introduction

In the spring of 2020 case counts doubled in under a week. A homogeneous compartmental model reads that growth as a reproduction number near 2.5 and a herd-immunity threshold of $1 - 1/R = 60\%$. First waves turned over with 20% or less of the population infected. Cotton (2020a) called this the herd-immunity paradox and argued that it follows from variation in transmission across people, places and time: early growth is dominated by the fastest transmitters, and the turnover by a harmonic mean (Cotton, 2020b).

Heterogeneity lowers the threshold because the most active people are infected first. They are also the people who would have passed the infection on most, so each infection removes more than its share of future transmission. Britton et al. (2020) and Gomes et al. (2022) made this quantitative for persistent differences in activity or susceptibility. Tkachenko et al. (2021) added activity that changes over time and showed that the resulting suppression is partly transient.

The strength of the effect is one number, the *immunity factor* λ , and a contact panel measures it directly. The panel separates activity into a part that persists for years, a part that comes and goes over weeks, and day-to-day noise. Only the first two matter, and together they predict $\lambda \approx 3.5$ with no epidemic data used.

That value matches what epidemics show wherever contacts were not cut by law or by a large voluntary drop. Where they were cut, first waves turned over with 1–5% infected, and the cut, not immunity, explains the turnover.

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2 The turnover rule

Unequal activity. Let S be the susceptible share. In a homogeneous population the effective reproduction number is $R_e = RS$. When people differ in activity, write

$$R_e = RS^\lambda. \quad (1)$$

Early on, $R_e \approx R(1 - \lambda(1 - S))$, so each percentage point of infection lowers R_e by λ percentage points of R . Setting $R_e = 1$ gives the infected share at the turnover,

$$a^* = 1 - R^{-1/\lambda}. \quad (2)$$

The textbook threshold is the case $\lambda = 1$. Equivalently, the textbook formula applies with the adjusted reproduction number $R^{1/\lambda}$.

Gamma activity. Let activity a have a gamma distribution with mean one and squared coefficient of variation $\text{CV}^2 = 1/k$, and let it scale both the chance of infection and the number of people infected. After cumulative force of infection Z , the susceptible share is $S = (1 + Z/k)^{-k}$ and $R_e \propto \mathbb{E}[a^2 e^{-aZ}] \propto (1 + Z/k)^{-k-2}$. Hence (1) holds exactly with

$$\lambda = 1 + 2\text{CV}^2, \quad (3)$$

as in Gomes et al. (2022).

Age structure. Let M_{ij} be the mean number of daily contacts of a person in age group i with people in group j , and n_i the population shares. Infected fractions grow along the leading right eigenvector v of M ; let u be the left eigenvector. Depleting susceptibles along v lowers the leading eigenvalue at the rate

$$\lambda_{\text{age}} = \frac{\sum_i u_i v_i^2 / \sum_i u_i v_i}{\sum_i n_i v_i}. \quad (4)$$

The numerator measures how strongly early infections sit in the groups that drive transmission; the denominator converts to the overall infected share. The same formula applies to any finite type structure, and we use it for age crossed with activity.

3 Activity that changes over time

A solvable model. Let each person's activity take two values with long-run shares p_1, p_2 , mean one and variance σ^2 , so that $(a - 1)^2 = c(a - 1) + \sigma^2$ for a constant c that sets the skew. Activity switches on a Markov clock with generator Q and relaxes at rate κ . A person with activity a is infected at rate $\beta a \Phi$, where $\Phi = \sum_k a_k I_k$, and infectious people recover at rate γ .

Proposition 1 (Growth, reproduction number and immunity factor). *The early growth rate r and the basic reproduction number solve*

$$1 = \beta \left(\frac{1}{r + \gamma} + \frac{\sigma^2}{r + \gamma + \kappa} \right), \quad R_0 = \beta \left(\frac{1}{\gamma} + \frac{\sigma^2}{\gamma + \kappa} \right). \quad (5)$$

During growth, with $\rho = r/(r + \kappa)$, the immunity factor is

$$\lambda = \frac{(1 + \sigma^2 \rho)/\gamma + \sigma^2 (1 + (1 + c)\rho)/(\gamma + \kappa)}{1/\gamma + \sigma^2/(\gamma + \kappa)}. \quad (6)$$

The proof is in Appendix A. The resolvent $(r + \gamma - Q^\top)^{-1}$ acts on only two directions, the stationary law p and the deviation $(a - 1) \circ p$, which decays at rate κ . Busy people infect busy people, and the second term in (5) is that correlation, kept for as long as a busy phase outlasts the infection.

Limits. With $\kappa = 0$, (6) is $\langle a^3 \rangle / \langle a^2 \rangle$, the value for persistent heterogeneity. As $\kappa \rightarrow \infty$ it tends to one. For fast switching,

$$\lambda - 1 \approx r K_a, \quad K_a = \frac{\sigma^2}{\kappa}, \quad (7)$$

where K_a is the Green–Kubo number of activity, the integral of its autocovariance. A fast epidemic outruns the reshuffling of activity and uses up the busy phase before it is refilled.

Effective variance. For the prediction in Section 5 we combine a persistent variance σ_p^2 with a switching variance σ_s^2 of correlation time τ through

$$\text{CV}_{\text{eff}}^2 = \sigma_p^2 + \frac{r}{r + 1/\tau} \sigma_s^2, \quad (8)$$

the persistent part acting in full and the switching part in proportion to the share of a busy phase that survives one growth time. This interpolates between the limits of Proposition 1 and is an approximation.

4 Mixtures of places

Cotton (2020b) reaches the same conclusion from a different direction. Divide a population into many places, each a standard epidemic with transmission rate $\beta = \bar{\beta} \eta$. The questions that matter are symmetric in the places, so by de Finetti’s theorem it suffices to consider mixtures of independent places. Early growth reads a transmission rate $\beta_g = t^{-1} \log \mathbb{E} e^{t\bar{\beta}\eta} = G \bar{\beta}$ with $G > 1$, and the susceptible share at the turnover is $\mathbb{E}[\gamma/\beta] = J/\bar{R}$ with $J = \mathbb{E}[1/\eta] \geq 1$. Together,

$$s^* \approx \frac{J G}{R_g}, \quad (9)$$

larger than the textbook $1/R_g$ by two convexity adjustments. When each place’s rate switches between $\bar{\beta} \pm \delta$ at rate ν , the total grows at the top eigenvalue $\bar{\beta} - \gamma - \nu + \sqrt{\nu^2 + \delta^2}$, which is $\bar{\beta} - \gamma + \delta^2/(2\nu)$ to first order: the growth convexity is again a Green–Kubo number.

5 The immunity factor from contact surveys

One-day diaries. The POLYMOD diaries (Mossong et al., 2008) record every contact of 7,290 people in eight European countries on one day. Within age groups the squared coefficient of variation of the number of contacts is 0.53 for ages 0–19, 0.63 for 20–64 and 0.77 for 65 and over. Combining these with the age matrices of Prem et al. (2021) through (4), with activity held fixed, gives $\lambda = 2.70$ for the United States, 2.77 for Sweden, 2.66 for Spain, 3.02 for the United Kingdom and 2.30 for Brazil. Age alone gives 1.3–1.7.

A one-day diary cannot tell a busy person from a busy day, so these values mix lasting and passing variation.

A two-year panel. The CoMix panel (Jarvis et al., 2020) surveyed the same people in the United Kingdom from March 2020 to March 2022. We use 21,158 adults aged 18–69 seen at least three times, 160,869 survey days in all. Activity is the number of contacts on the diary day divided by the mean for the same week and age group, so lockdowns and seasons drop out. The covariance of a person’s activity between surveys ℓ days apart is fitted as

$$C(\ell) = \sigma_p^2 + \sigma_s^2 e^{-\ell/\tau} \quad (\ell \geq 7), \quad C(0) = \sigma_p^2 + \sigma_s^2 + \sigma_n^2, \quad (10)$$

with a persistent part, a switching part of correlation time τ , and day-to-day variation σ_n^2 .

contacts capped at	one-day CV^2	persistent σ_p^2	switching σ_s^2	τ (days)	day to day σ_n^2
20	1.82	0.54 (30%)	0.23 (13%)	36	1.05 (57%)
50	3.90	1.06 (27%)	0.51 (13%)	28	2.34 (60%)
100	6.54	1.69 (26%)	0.88 (14%)	23	3.97 (61%)

Table 1: Decomposition of adult activity in the CoMix UK panel.

The split in Table 1 is stable across caps. About 60% of one-day variation is gone within a week and does not change who an epidemic uses up. About 27–30% persists for two years. About 13% comes and goes with a correlation time of three to five weeks.

Prediction. Inserting (8) into the joint age and activity structure with the UK matrix gives the immunity factors in Table 2, for growth rates from 0.02 to 0.2 per day.

contacts capped at	λ	turnover at $R = 1.6$	turnover at $R = 2.5$
20	3.4–3.7	12–13%	22–24%
50	4.6–5.2	9–10%	16–18%
100	5.9–6.8	7–8%	13–14%

Table 2: Immunity factor predicted from the CoMix decomposition and the UK age matrix, with no epidemic data.

The immunity factor changes by only about 7% across a tenfold range of growth rates, because the switching part is small. The level depends on how the few very high contact counts are treated. The right cap depends on how many of a very busy person’s contacts can transmit, and we report three.

6 The immunity factor in epidemics

US county waves. We identify 471 epidemic waves in 372 US counties of more than 50,000 people from daily deaths (New York Times, 2021), February 2020 to June 2021. For each wave, early growth gives R through a gamma generation interval of mean 5.5 days and standard deviation 2.1 days (Wallinga and Lipsitch, 2007). Cumulative deaths divided by an infection fatality rate give the infected share when the wave began, a_0 , and at the death peak, a_1 . The immunity factor of the wave is

$$\lambda = \frac{\ln R}{-\ln((1 - a_1)/(1 - a_0))}. \quad (11)$$

Between 95% and 100% of waves in every season turned over below the homogeneous threshold. In winter 2020–21, with 341 waves, a constant immunity factor fits at 3.5 for a fatality rate of 0.7%, 2.4 at 0.5% and 5.3 at 1%.

The slope is not identified. Grouped by growth rate, the winter waves infected nearly the same share of their susceptibles whatever their R . Growth read from deaths and from cases three weeks earlier correlate at only 0.23 in winter, and noise in a regressor flattens any fitted slope. With case growth as an instrument the slope of log depletion on $\log \ln R$ is 0.91 in spring (90% interval 0.46 to 1.50) and 2.9 in winter (0.7 to 12.5). The county data identify the level of λ but not its dependence on growth, which Table 2 says is small.

Serology. Antibody surveys measure the infected share directly. Scaling each survey back to the death peak with cumulative deaths gives Table 3.

place	survey	contacts	R	infected	λ
England, 9 regions	Ward et al. (2021)	lockdown	2.1–3.5	0.9–5.0%	17–94
Geneva	Stringhini et al. (2020)	closures	3.3	5.1%	23
Stockholm	Castro Dopico et al. (2021)	voluntary	2.8	2.3–3.1%	33–44
Spain, spring	Pollán et al. (2020)	lockdown	1.9–4.3	0.3–4.6%	about 60
Spain, autumn	round 4	curfew	1.1–1.5	2.6% [†]	about 7
Manaus	Buss et al. (2021)	little	1.6	about 17%	2.5

Table 3: Turnovers anchored by serology; the infected share is at the turnover. [†]Share of those still susceptible after the first wave.

Where contacts were cut, by law or voluntarily, the first wave turned over with 1–5% infected and λ exceeds 15. The cut in contacts, not immunity, accounts for these turnovers. Where control was weak, in Manaus and in the US county waves of winter 2020–21, λ is 2 to 3.5. Tkachenko et al. (2021) fit about 4 to New York City and Chicago.

7 Discussion

The immunity factor predicted from contact surveys, 3.4–3.7, sits inside the range measured from epidemics where contacts were not cut, 2 to 4. At $R = 2.5$ it puts the turnover at 22–24%, which is the paradox as first observed.

On the contact side the level depends on how very high contact counts are treated. On the epidemic side it scales with the assumed fatality rate. The two uncertainties are of similar size.

Most of the heterogeneity that matters persists. The switching part is small and wears off over about a month, so a first-wave turnover is mostly a lasting reduction of the threshold with a transient on top. This sits uneasily with the 66–76% attack reported for Manaus, which persistent heterogeneity of this size would hold well below. Either heterogeneity was smaller there or the blood-donor serology overstates the attack.

The prediction can be tested out of sample. CoMix panels exist for about twenty countries. Fixing the treatment of high contact counts in advance, each panel predicts an immunity factor that the country’s serology and death curves can confirm or refute.

A Proof of Proposition 1

Let $I = (I_1, I_2)$ be the infectious people by current activity and $\Phi = a^\top I$. Early on, $\dot{I} = \beta \Phi (a \circ p) - \gamma I + Q^\top I$. Seek $I = v e^{rt}$:

$$v = \beta \Phi (r + \gamma - Q^\top)^{-1} (a \circ p). \quad (12)$$

Since $Q^\top p = 0$ and $Q^\top ((a-1) \circ p) = -\kappa (a-1) \circ p$ for a two-state chain, and $a \circ p = p + (a-1) \circ p$,

$$(r + \gamma - Q^\top)^{-1} (a \circ p) = \frac{p}{r + \gamma} + \frac{(a-1) \circ p}{r + \gamma + \kappa}. \quad (13)$$

Taking $a^\top$ of both sides, with $a^\top p = 1$ and $a^\top ((a-1) \circ p) = \sigma^2$, gives (5); R_0 is the case $r = 0$.

For the immunity factor, the susceptible deficit $s = p - S$ solves $\dot{s} = \beta \Phi (a \circ S) + Q^\top s$, and during growth $s = \beta \Phi (p/r + (a-1) \circ p/(r + \kappa))$ to first order, with overall attack $x = \beta \Phi / r$. Then $a \circ s$ has components $x(1 + \sigma^2 \rho)$ along p and $x(1 + (1 + c)\rho)$ along $(a-1) \circ p$, using $(a-1)^2 = c(a-1) + \sigma^2$. The next-generation number with susceptibles S is $\beta a^\top (\gamma - Q^\top)^{-1} (a \circ S)$, and subtracting its value at $S = p$ gives $R_e = R_0 (1 - \lambda x)$ with λ as in (6). The formulas agree with numerical solution of the model: the growth rate to four digits and λ to 0.1–0.3% (04_activity_switching.py).

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